

VOXAI SQL

(RESEARCH AND DEVELOPMENT)

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Chapter 1:

Introduction

In today's data-centric world, databases are an essential component for managing and storing information. They support a wide range of applications in business, research, and personal data management. However, interacting with databases often requires specialized knowledge in Structured Query Language (SQL), making it challenging for non-technical users to access and manage data effectively. This complexity creates a barrier to efficient data management, limiting access to critical information and making data retrieval cumbersome.

Despite advancements in user interfaces and the development of voice-controlled systems, these solutions still rely heavily on exact SQL syntax or rigid command templates, rendering them difficult to use for most non-technical users. Furthermore, voice-based systems often struggle with errors in speech recognition, which can result in incorrect queries and frustration. To address these challenges, new developments in natural language processing (NLP) and neural networks offer the potential to make database interactions more intuitive and accessible.

This research proposes a novel system—**VoxAI SQL**—which aims to simplify database management by enabling users to interact with databases through natural language and voice commands, eliminating the need for SQL expertise. By leveraging state-of-the-art NLP models and integrating voice recognition capabilities, VoxAI SQL transforms complex database queries into a seamless, conversational experience, making databases more accessible for non-technical users and people with physical constraints like Repetitive Stress Injury (RSI).

1.1 Problem Domain

Working with databases is a crucial skill in the digital era, yet it often requires a strong understanding of SQL (Structured Query Language). This complexity makes data management and retrieval challenging for non-technical users, who may not have the necessary technical skills. Modern database management systems offer user-friendly interfaces, but these are usually still limited in handling advanced queries, requiring a deep understanding of database structures and SQL syntax.

Voice-controlled systems have been introduced as an alternative to simplify interactions. However, they typically require precise SQL commands or predefined templates, which can be daunting for non-technical users. Moreover, speech recognition errors further complicate these interactions, leading to incorrect data retrieval and reduced user satisfaction. Consequently, there is a pressing need for a system that allows users to interact naturally with databases using everyday language, bypassing the need for SQL knowledge and minimizing the impact of speech recognition inaccuracies.

1.2 Research Problem Statement

The main research problem addressed in this project is the lack of intuitive, accessible tools that allow non-technical users to interact with databases using natural language or voice commands. Managing databases typically requires knowledge of SQL, creating a significant barrier for non-technical users. Current voice-based

database systems are not user-friendly as they still rely on rigid command structures and strict SQL syntax, which demand some degree of technical understanding. Furthermore, these systems are prone to errors caused by inaccurate speech recognition, which can result in invalid queries.

This project proposes the development of an AI-powered chatbot capable of handling natural language queries to overcome these challenges. By converting voice and text commands into SQL queries, the system will allow users to perform both simple and complex database operations without requiring SQL expertise. The solution aims to bridge the gap between technical and non-technical users, making database management more accessible and reducing the reliance on traditional interfaces.

1.3 Software Requirements Specifications

This section outlines the functional and non-functional requirements necessary for developing the proposed AI chatbot system. These requirements define the core features and constraints to ensure a successful implementation that meets the project's goals.

1.3.1 Functional Requirements:

1. Natural Language Processing (NLP) Module:

- The system shall include an NLP module capable of interpreting user queries and converting them into appropriate database operations.
- It shall support text-based and voice-based inputs for flexibility.

2. Voice Recognition Module:

- The system shall include a voice recognition module that accurately transcribes spoken language into text commands for the NLP module.
- It should handle variations in speech patterns and accents with minimal errors.

3. SQL Query Generator:

- The SQL generator shall convert interpreted natural language into corresponding SQL queries based on user input.
- It should support complex SQL operations such as joins, aggregations, and nested queries.

4. Database Connectivity:

- The system shall connect to MySQL databases to execute the generated SQL commands.
- It should support secure, real-time interaction with the database to provide immediate results.

5. User Feedback Module:

- The system shall provide feedback to the user on the generated queries, explaining the executed commands for transparency and learning purposes.

1.3.2 Non-Functional Requirements:

1. User-Friendly Interface:

- The system shall have a simple, intuitive interface with multi-language support to accommodate diverse users.

2. High Accuracy:

- The NLP and voice recognition modules shall achieve high accuracy in interpreting user inputs and converting them into valid queries.

3. Real-Time Response:

- The system shall respond to user inputs in real time to provide a seamless interaction experience.

4. Scalability:

- The system's architecture shall be scalable to support larger databases and more complex queries in the future.

5. Data Security:

- The system shall ensure the confidentiality and integrity of user data through encryption and secure authentication method.

Chapter 2

Literature Review

In recent years, the evolution of voice-based interactions has dramatically transformed the way users interact with technology. As voice assistants such as Siri, Alexa, and Google Assistant have become ubiquitous, there is growing interest in expanding voice-controlled systems to more complex applications, including database management. While traditional database systems rely heavily on Structured Query Language (SQL) for data manipulation, the learning curve associated with SQL syntax and the technical expertise required to execute complex queries pose significant challenges for non-technical users. As a result, various research efforts have focused on developing natural language processing (NLP)-driven models to bridge the gap between technical and non-technical users, enabling them to interact with databases using everyday language or voice commands. Despite advancements, many existing solutions lack the flexibility and accuracy needed to handle real-world scenarios effectively.

This chapter critically examines the current state of the art in voice-to-SQL query generation, focusing on the strengths, limitations, and potential areas of improvement. We analyze key research contributions, their methodologies, and their relevance to this project, identifying how they inform the development of a more robust, AI-driven solution that addresses current gaps.

2.1 Related Research

2.1.1 “AI Model to Generate SQL Queries from Natural Language Instructions through Voice” (Sawant, 2022) **Summary of the Research Item**

The paper presents a web-based system that simplifies database interactions for non-technical users by converting voice commands into SQL queries using a Long Short-Term Memory (LSTM) network. The proposed system involves several steps: converting speech to text, analyzing the text through NLP techniques, and generating corresponding SQL queries. The goal is to make data management more accessible by eliminating the need for SQL proficiency.

Critical Analysis

This research is effective in reducing the complexity of basic SQL queries, making it user-friendly for non-technical audiences. However, the model struggles with complex query structures, limiting its application in

scenarios requiring multi-table operations or nested SQL queries. Additionally, its reliance on static speech-to-text modules introduces potential for transcription errors, impacting the final SQL output.

Relationship to the Proposed Research Work

VoxAI SQL builds on Sawant’s foundational work by incorporating Transformer-based models for improved accuracy in complex SQL query generation. Although the current system lacks a voice-to-text module, this feature will be added in future iterations to support a fully voice-driven interface.

2.1.2 “Speech-to-SQL: Toward Speech-Driven SQL Query Generation from Natural Language Questions” (Song, 2024)

Summary of the Research Item

The study introduces SpeechSQLNet, a model that converts spoken language directly into SQL commands using a dual-attention mechanism. Unlike traditional systems that rely on speech-to-text conversion, SpeechSQLNet maps spoken commands directly to SQL structures. The model was evaluated using a custom dataset called SpeechQL, achieving better performance than conventional text-based methods.

Critical Analysis

The novelty of SpeechSQLNet lies in bypassing the speech-to-text stage, reducing the risk of transcription errors. However, its effectiveness is limited when dealing with complex SQL operations that require deeper semantic understanding. Additionally, its use of synthetic datasets may not accurately reflect real-world speech, making it less reliable in diverse settings.

Relationship to the Proposed Research Work

VoxAI SQL aims to extend this approach by integrating context-aware models to handle complex SQL queries. While our system currently lacks direct speech-to-SQL capabilities, future development will incorporate dual-attention mechanisms similar to those in SpeechSQLNet for improved accuracy in voice-driven query generation.

2.1.3 “SpeakQL: Towards Speech-driven Multimodal Querying of Structured Data” (Vraj Shah, 2020)

Summary of the Research Item

The paper proposes a multimodal system that combines voice and touch-based interactions for SQL generation. Users can refine SQL queries through voice commands and manually adjust them using visual feedback. This human-in-the-loop approach allows users to see how spoken commands translate into SQL, making it easier to learn and understand the process.

Critical Analysis

The multimodal interaction method is effective for user engagement and learning. However, the system's dependence on manual corrections for complex queries reduces its automation potential. Furthermore, the need for visual feedback limits its use in hands-free environments, making it less suitable for voice-only interactions.

Relationship to the Proposed Research Work

VoxAI SQL will draw on SpeakQL's multimodal approach to provide visual feedback for SQL generation but aims to minimize the need for manual corrections. Our focus will be on automating complex query generation through advanced NLP models, making the system more suited for voice-only environments.

2.1.4 “VoiceQuerySystem: A Voice-driven Database Querying System Using Natural Language Questions” (Yuanfeng Song, 2022)

Summary of the Research Item

VoiceQuerySystem converts natural language questions into SQL commands using two distinct methods: a traditional speech-to-text approach followed by text-to-SQL conversion and a direct speech-to-SQL method using SpeechSQLNet. The system uses a specialized dataset called SpeechQL to evaluate its performance and has shown promising results for basic SQL queries.

Critical Analysis

The system's flexibility in supporting multiple approaches is a notable strength. However, its dependency on the ASR (Automatic Speech Recognition) system introduces vulnerabilities, as transcription errors can affect the SQL generation process. Furthermore, the use of a limited dataset restricts its adaptability to real-world scenarios with diverse accents and speech patterns.

Relationship to the Proposed Research Work

This research highlights the importance of robust ASR systems, which will be a key consideration in VoxAI SQL’s development. While the current system lacks direct speech-to-SQL integration, future iterations will address these limitations by incorporating context-aware models that can adapt to diverse speech inputs.

2.1.5 “TalkSQL: A Tool for the Synthesis of SQL Queries from Verbal Specifications” (Obaido, 2020)

Summary of the Research Item

TalkSQL enables users to create SQL queries through spoken instructions using a multi-step process involving speech-to-text conversion, NLP analysis, and SQL synthesis. The system focuses on providing accessibility for people with disabilities and non-technical users, offering clear feedback on the generated SQL commands.

Critical Analysis

While TalkSQL is effective for basic SQL operations, it struggles with complex queries due to its reliance on pattern matching and regular expressions. This limits its flexibility in handling more sophisticated SQL commands and complex database schemas.

Relationship to the Proposed Research Work

VoxAI SQL will incorporate some of TalkSQL’s accessibility features, such as feedback mechanisms for user learning, but will employ more advanced NLP models to handle complex SQL operations, making the system more versatile for real-world applications.

2.1.6 “CHAT-SQL: Natural Language Text to SQL Queries Based on Deep Learning Techniques” (MAJHADI Khadija, 2024)

Summary of the Research Item

CHAT-SQL explores the challenges of converting natural language into SQL using deep learning models. The study uses LSTM networks to translate natural language queries into SQL commands, addressing the syntactic and semantic variations that arise in natural language inputs. The system was evaluated on complex queries, achieving an accuracy of 89%.

Critical Analysis

The LSTM-based approach shows high accuracy for standard SQL queries but struggles with ambiguous language inputs. The model's performance is dependent on the quality and variety of the training data, making it less effective in handling diverse real-world scenarios.

Relationship to the Proposed Research Work

VoxAI SQL will build on CHAT-SQL's findings by employing more sophisticated models, such as Transformers, to better handle ambiguous queries and improve contextual understanding.

2.1.7 “Natural Language to SQL: Automated Query Formation Using NLP Techniques” (Sri Lalitha Y.*, 2023) Summary of the Research Item

The paper addresses the complexities of SQL query generation using natural language inputs. It employs various NLP techniques, such as parsing and synonym mapping, to convert English text into SQL commands. The system is designed to assist non-technical users by allowing them to formulate queries in plain English without worrying about SQL syntax.

Critical Analysis

The system performs well for straightforward queries but struggles with more complex structures involving nested clauses. Additionally, it lacks support for contextual understanding, which is crucial for interpreting user intent in dynamic environments.

Relationship to the Proposed Research Work

VoxAI SQL aims to overcome these limitations by implementing context-aware models that can handle complex queries and provide better intent detection.

2.1.8 “Database-Aware ASR Error Correction for Speech-to-SQL Parsing” (Shao, Kumar, & Nakashole, 2023) Summary of the Research Item

The study introduces DBATI, an error-correction mechanism for ASR outputs in speech-to-SQL systems. DBATI uses a combination of Tagger and Infilling Language Model (ILM) to identify and correct transcription errors based on the database schema, leading to more accurate SQL query generation.

Critical Analysis

DBATI significantly improves SQL accuracy by considering the database schema context. However, the joint encoding approach increases model complexity and may not be suitable for real-time applications due to high computational requirements.

Relationship to the Proposed Research Work

VoxAI SQL will incorporate similar error-correction mechanisms to improve the robustness of its voice-driven SQL generation, ensuring that transcription errors do not compromise the final output.

2.2 Analysis Summary of research items

<i>Research Paper Name</i>	<i>Publish Year</i>	<i>Models Used</i>	<i>Accuracy</i>	<i>Dataset</i>	<i>Strengths</i>	<i>Weakness</i>
“CHAT-SQL: NATURAL LANGUAGE TEXT TO SQL QUERIES BASED ON DEEP LEARNING TECHNIQUES ” (MAJHADI Khadija, 2024)	2024	LSTM-based neural network	89 %	Spider dataset	□ Comprehensive analysis of sequence-to-sequence models for NL to SQL conversion. □ Proposes an innovative model with strong performance (89% accuracy) for complex queries. □ Evaluates using multiple metrics, providing a thorough performance assessment. □ Practical focus on developing a mobile application for better user interaction.	□ Limited evaluation of datasets like WikiSQL and Spider without proposing solutions. □ Insufficient exploration of advanced methods for resolving NL ambiguities. □ Lacks real-world testing to validate the model’s practical effectiveness. □ Scalability concerns with larger or more complex datasets are not addressed.
“Speech-to-SQL: toward speech-driven SQL query generation from natural language question” (Song, 2024)	2024	SpeechSQL Net, SpeechSeq2SQL	54.70%	SpeechQL WikiSQL Spider	□ SpeechSQLNet directly translates speech to SQL, reducing errors. □ Pre-training techniques enhance performance by bridging speech-text gaps. □ Resource-efficient design suits mobile and low-power environments. □ Schema encoders improve accuracy in linking speech to database schemas.	□ Speech recognition may struggle with diverse accents. □ Reliance on SpeechQL limits generalizability. □ Pre-training models need further development for better performance.
“Natural Language to SQL: Automated Query Formation	2023	Multinomial Naive Bayes, Customized (NER) Model,	92.31%	Qualetics Company dataset	□ User-friendly for non-technical users, eliminating the need to learn SQL. □ Speeds up data retrieval by generating	□ Struggles with complex SQL queries involving joins or constraints.

Using NLP Techniques” (Sri Lalitha Y.*, 2023)		SVM, SMOTE			SQL queries from natural language. □ Dynamically creates SQL queries from English text inputs. □ Includes speech-to-text for easier query input.	□ Depends on NLP techniques, which may miss language nuances. □ Ambiguous queries can lead to misinterpretation. □ Needs improvements to handle complex queries. □ Performance depends on the quality and diversity of the training data.
“DATABASE-AWARE ASR ERROR CORRECTION FOR SPEECH-TO-SQL PARSING” (Shao, Kumar, & Nakashole, 2023)	2023	LSTM-CRF, Infilling Language Model (ILM) Rewriter, Pre-trained Language Model (PLM)	59.50%	Spider	□ DBATI combines Tagger and ILM for better SQL query accuracy. □ Significant SQL accuracy improvements with WER and BLEU metrics. □ Useful for healthcare database interactions. □ Spoken Spider dataset aids future research.	□ Results may not generalize to other ASR systems or parsing methods. □ Effectiveness depends on ASR output quality; errors can impact accuracy. □ Controlled datasets used; real-world testing is needed. □ Joint encoder adds complexity, potentially challenging in resource-constrained settings.
“AI Model to Generate SQL Queries from Natural Language Instructions through Voice” (Sawant, 2022)	2022	LSTM	93.47%	Json formatted dataset	□ LSTM models are well-suited for SQL query generation from natural language. □ Voice commands enhance usability for users with RSI or those preferring voice interaction. □ Effective data preprocessing improves model accuracy.	□ The system struggles with highly complex SQL queries. □ Model accuracy is dependent on the quality and diversity of training data. □ Web-based performance may be affected by internet and browser issues.
“VoiceQuerySystem: A Voice-driven Database	2022	ASR + IRNet, SpeechSQL	53.95%	SpeechQL	□ The E2E approach in SpeechSQLNet processes speech	□ Cascaded methods introduce errors through

Querying System Using Natural Language Questions” (Yuanfeng Song, 2022)		Net, Seq2SQL			directly into SQL, enhancing accuracy. □ Advanced neural architecture and pre-training boost performance over traditional models. □ User-friendly design allows intuitive database interactions for those with limited SQL knowledge.	intermediate text, reducing reliability. □ The E2E model’s accuracy depends on ASR quality, which can affect SQL generation. □ The current system supports only pre-recorded audio, limiting real-time use.
“SpeakQL: Towards Speech-driven Multimodal Querying of Structured Data” (Vraj Shah, 2020)	2020	Automatic Speech Recognition (ASR), Azure's Custom Speech Service, Metaphone	64%	Yelp Dataset	□ SpeakQL supports both speech and touch corrections, enhancing user-friendliness. □ Human-in-the-loop mechanisms improve query accuracy, boosting it by up to 50%. □ SpeakQL significantly speeds up SQL query specification, with improvements from 2.7x to 6.7x. □ Targeted user research ensures the system meets real user needs across sectors.	□ SpeakQL may not support all SQL functionalities, limiting its use for complex queries. □ The system's effectiveness depends on ASR quality; poor transcription can still affect the user experience. □ Non-technical users may face a learning curve, limiting accessibility despite speech input.
“TalkSQL: A Tool for the Synthesis of SQL Queries from Verbal Specifications” (Obaido, 2020)	2020	Hidden Markov Models, Regular Expressions (RE)	Higher Accuracy	Sample database schema	□ TalkSQL simplifies database queries through voice commands, aiding users like visually impaired learners. □ Automated feedback helps users quickly learn and correct SQL query mistakes. □ High usability ratings show that most	□ Complex queries, especially those with nested statements, are struggled with by TalkSQL. □ The ability to handle more SQL commands is limited by its reliance on regular expressions. □ All user intents might not be covered by the feedback, which can

					participants found TalkSQL easy to use for CRUD commands. ☐ The tool has potential for educational applications, offering an interactive way to learn SQL.	lead to misunderstandings. ☐ Better handling of complex queries requires advanced parsing techniques
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2.3 Research Gap

The reviewed research papers indicate a significant gap in providing a robust, real-world applicable solution for voice-driven SQL query generation. Current models are limited in handling complex queries and are prone to errors caused by speech recognition inaccuracies. Additionally, the dependency on structured datasets for training restricts their ability to adapt to dynamic, real-world scenarios. This research aims to bridge these gaps by developing a more flexible, context-aware model that can accurately interpret natural language commands, handle complex SQL operations, and provide a seamless voice-based interface for non-technical users.

Chapter 3

Proposed Approach

This chapter outlines the proposed methodology and architectural design of the **VoxAI SQL** system. The goal of the proposed approach is to enable users to interact with databases using natural language and voice commands, thereby eliminating the need for traditional SQL knowledge. The system will leverage state-of-the-art AI models to convert voice inputs into structured SQL queries and execute them on the specified databases. The architecture is designed to be modular, allowing for the integration of new components such as advanced NLP models and voice processing frameworks.

3.1 Overview of the System Architecture

The proposed system is divided into three primary components: **Frontend**, **Backend**, and **AI Models**. Each component is designed to handle specific tasks, ensuring smooth data flow and accurate query generation. Figure 3.1 below presents the architectural diagram of the system, which outlines the interactions between the frontend, backend, and AI models.

3.1.1 Frontend (Next.js)

The frontend is built using Next.js and serves as the user interface for all interactions. It includes the following key pages:

- 1. **Login Page**: Allows users to log in using their credentials.

2. **Register Page:** Provides a form for new users to register and create accounts.
3. **Chat Page:** The core interaction page where users can input text commands or use voice commands to communicate with the system. The chat interface allows users to send queries and view results in real-time.
4. **Dashboard:** Displays a summary of chat history and results retrieved from the database.

The frontend communicates with the backend by sending user commands, which are then processed and routed to the appropriate AI models.

3.1.2 Backend (Express.js)

The backend is implemented using the Express.js framework and serves as the central controller for handling API requests, managing database operations, and interacting with the AI models. It includes the following routes and controllers:

- **AuthRoute & AuthController:** Handles user authentication, including login and registration.
- **UserRoute & UserController:** Manages user profiles and permissions.
- **DatabaseRoute & DatabaseController:** Facilitates database management, allowing users to add, update, or delete databases.
- **ChatRoute & ChatController:** The core route for processing chat commands. It sends user inputs to the appropriate AI models for voice-to-text conversion, SQL generation, and result retrieval.

3.1.3 AI Models

The AI models are the core of the system, responsible for interpreting natural language inputs and generating corresponding SQL queries. The following models will be integrated into the system:

1. **Voice-to-Text Model:** The whisper.cpp model will be used for converting user voice inputs into text. Whisper is a state-of-the-art speech recognition model that provides accurate transcription for diverse languages and accents. By integrating this library, the system will be able to handle a variety of speech inputs with minimal errors.
2. **SQL Generation Model:** The SQL generation model will be developed in-house using a Transformer-based architecture. The model will take natural language text (converted from voice commands) as input

and generate structured SQL queries as output. This model will handle both simple queries (e.g., single-table retrieval) and complex queries (e.g., nested joins and aggregations).

3.1.4 Schemas and Data Storage

The system will maintain separate database schemas for storing user profiles, chat records, database information, and SQL query logs. The schemas are defined as follows:

- **UserSchema:** Stores user registration information, authentication details, and user roles.
- **DatabaseSchema:** Keeps track of databases connected by the user, including metadata and permissions.
- **ChatSchema:** Logs user interactions, including queries, SQL commands, and results.
- **MessageSchema:** Stores the details of each message exchanged between the user and the system, allowing for chat history retrieval.

3.2 Methodology

The methodology for developing **VoxAI SQL** involves several stages, each targeting a specific aspect of the system's functionality:

3.2.1 Stage 1: Voice-to-Text Conversion

- **Tool:** whisper.cpp
- **Description:** The Whisper model will be used to convert user voice inputs into text. Whisper has been chosen for its robust handling of diverse accents and its ability to transcribe complex speech patterns accurately. It will serve as the foundation for building a reliable voice-driven interface.
- **Integration:** The transcribed text will be sent to the backend, where it will be processed by the ChatController before being passed to the SQL Generation Model.

3.2.2 Stage 2: Natural Language to SQL Conversion

- **Tool:** Custom-built Transformer-based model
- **Description:** The natural language-to-SQL conversion model will be designed to handle both basic and complex SQL queries. The model will be trained on large-scale text-to-SQL datasets to ensure that it understands a wide variety of query structures and can generate accurate SQL statements based on user intent.

- **Process:**

1. The model will take natural language text as input and parse it to identify keywords and query components.
2. Using context-aware layers, it will map the input text to SQL structures.
3. The generated SQL query will be validated against the database schema to ensure correctness before execution.

3.2.3 Stage 3: SQL Execution and Response Handling

- **Description:** The validated SQL query will be executed against the connected database using the DatabaseController. The resulting data will be formatted and sent back to the frontend, where it will be displayed to the user in a readable format.

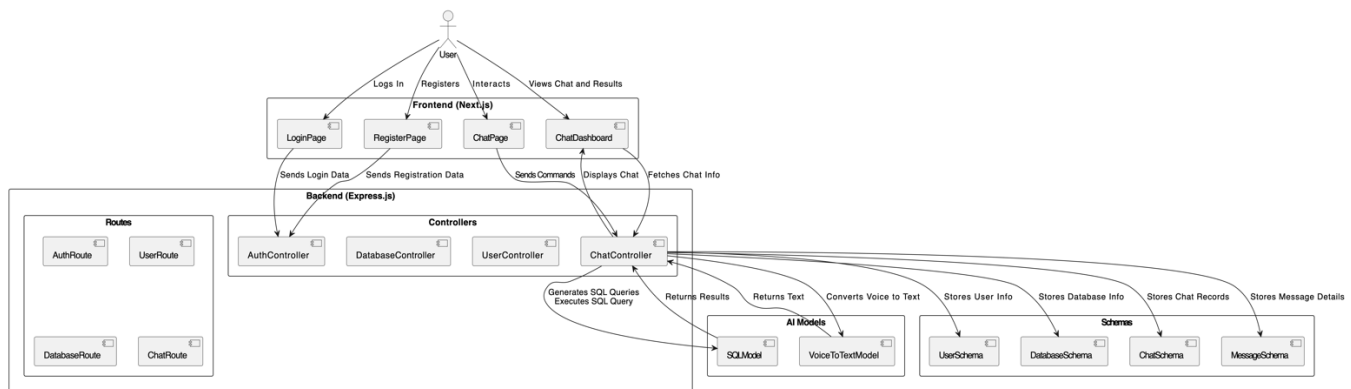
3.2.4 Placeholder for Future AI Models

In future iterations, the system will include text-to-voice modules to convert the results back into spoken form, enabling a fully hands-free experience. The placeholder in the architectural diagram represents the space allocated for these modules, ensuring that the design remains scalable and modular.

3.3 Graphical Representation

Figure 3.1 below illustrates the high-level architecture of the system, showcasing the flow of data between the frontend, backend, and AI models. The placeholder sections indicate where future models, such as Text-to-Speech, will be integrated.

Figure 3.1: Architectural Diagram of VoxAI SQL System.



3.4 Advantages of the Proposed Approach

The proposed approach offers several key benefits:

- **Modularity:** Each component is designed to be modular, allowing for easy updates and the integration of new models.
- **Scalability:** The architecture can handle a growing number of users and databases without compromising performance.
- **Flexibility:** By incorporating state-of-the-art AI models, the system can adapt to various input types and complex query structures.
- **User-Centric Design:** The integration of visual feedback and voice interaction ensures a user-friendly experience, catering to non-technical users.

3.5 Challenges and Future Directions

While the current design addresses the core challenges of voice-based SQL generation, future work will focus on enhancing the model's ability to understand context and handle ambiguous queries. The implementation of Text-to-Speech modules and advanced voice recognition algorithms will further refine the user experience, providing a more seamless interaction with databases.

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